

- Q4 A. When a device interrupt occurs, how does the processor determine which device issued the interrupt? 06
- B. What is the difference between programmed-driven I/O and interrupt-driven I/O? Give one advantage and one disadvantage of each. 06
- C. List three broad classifications of external, or peripheral, devices. 06
- Q5. A. Describe the following terminologies associated with pipeline computers: 06
- (i) Static Pipeline
 - (ii) Minimum Average Latency
 - (iii) Multifunctional Pipeline
 - (iv) Forbidden Latencies
 - (v) Collision Vector
 - (vi) Systolic Array
- B. Draw the interconnection structures and data flow patterns in general pipelines with either feedforward or feedback connections, in addition to the cascaded connections in a linear pipeline. Assume that the pipeline is dual functional, denoted as function A and function B. Also number the pipeline stages S1, S2, and S3 from the input end to the output end. Use the reservation table to show the utilization of successive pipeline stages. 06
- C. Classify the pipeline processor according to the: 06
- (i) Handler
 - (ii) Ramamoorthy and Li